

Cluster Autoscaler (CAS) in AKS

- **Cluster autoscaler automatically adjusts the number of nodes** to run it in an effective and economic way
- **Scale-up**
 - It monitors the **unschedulable pods due to resource constraints** and increases the number of nodes to meet application demand
 - The **default interval** for checking unschedulable pods is **10 seconds**, which **can be modified**
 - Cluster autoscaler attempts to find a new place to run unscheduled pods whenever **Kubernetes scheduler is unable to find a place** to operate them
- **Scale-down**
 - It also periodically assesses the number of nodes and **decreases** them as necessary as per:
 - Nodes are considered for removal if the **total amount of CPU and memory** used by all running pods is **less than 50%** of the node's available resources
 - **All running pods** except manifest-run pods or pods generated by DaemonSets **are transferred to other nodes** before considering the node for removal
 - Others
 - The removal of an unneeded node is configurable based on the **--scale-down-unneeded-time** flag
- A ConfigMap named cluster-autoscaler-status is added to the kube-system namespace to reflect the status
- HorizontalPodAutoscaler scales the number of pods, while Cluster Autoscaler scales the number of nodes. These two components can efficiently work together or individually